DecisionRL: Enhancing Decision Calibration through Reinforcement Learning

Reinforcement Learning Course Project Proposal

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Research Question

As large language models evolve from question answerers to decision makers, efficient and reliable decision-making becomes increasingly important. While System 2-style models emphasize deliberate, multi-step reasoning, System 1-style models aim to provide fast, intuitive judgments. Jev introduces a complementary paradigm of structured probabilistic decisions, producing explicit probability distributions over candidate actions instead of lengthy reasoning trajectories. Inspired by this paradigm, our project explores how **Reinforcement Learning for Calibrated Decisions (RLCD)** can enhance decision accuracy, probability calibration, and reliability through a reproducible training framework.

Related Work

Jev introduces System 1-style probabilistic decision-making and RLCD as an alternative to autoregressive reasoning. However, its model weights, training code, and full training recipe remain proprietary, limiting independent reproduction. **Laya** provides an open-source implementation of Jev-compatible typed decisions and RLCD, offering an accessible foundation for further research. **Sev** investigates RLCD optimization through independent reproduction and analysis. **Kev** and **Strands Decider** extend the open-source decision-model ecosystem through alternative architectures and training pipelines. Despite these advances, RLCD optimization remains relatively underexplored, particularly in reward design and policy optimization. Our project aims to develop a reproducible RLCD training framework and investigate improvements to its optimization.

Planned Approach

We develop DecisionRL to explore improved reinforcement learning strategies for reliable and calibrated decision-making. We explore ModernBERT and EuroBERT as pretrained encoder backbones.

- **Data Construction and Curation.** We use all official `typed-decisions` training examples and collect additional open-source decision datasets across multiple domains. Our data construction focuses on two types of supervision: hard labels for SFT and probabilistic calibration labels for RLCD. Since reliable probabilistic labels are scarce, we synthesize diverse decision scenarios and construct calibration labels from teacher-model logit distributions or aggregated predictions through repeated sampling and voting.
- **Model Training and RLCD Optimization.** We adopt a two-stage training framework. First, we perform supervised fine-tuning (SFT) on high-quality decision datasets to establish general decision-making capabilities. We then apply RLCD to optimize decision probabilities using verified probabilistic supervision. We also explore improvements to Laya's original RLCD algorithm through enhanced reinforcement learning optimization strategies for more reliable and calibrated decisions.

Evaluation Plan

We compare the original pretrained backbones, released Laya, SFT, SFT + RLCD, and DecisionRL on the held-out test set. We report **Accuracy**, **Macro F1 Score**, **Expected Calibration Error (ECE)**, and **Brier Score** to measure decision correctness, class-balanced performance, confidence calibration, and probability quality, respectively.

Required Resources and Division of Responsibilities

Resources: No additional equipment or specialized resources are required.

Preliminary Division of Responsibilities: Both team members will jointly participate in project discussions, research planning, and RLCD methodological design. Yanyun LIU will primarily contribute to data collection, synthesis, and quality control, while Taihua Chen will focus more on model and algorithm optimization.